CFM301 Project 3: Reproducing the Equity Return Predictability Table

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1. Economic question and audience

Question: is equity return predictability still present after 2011, and should a long-horizon investor adjust equity exposure based on valuation ratios? We reproduce the predictability table of Chapter 8 (slides 24-25 and the surrounding blocks): correlations and Newey-West t-statistics between each predictor's level today and future equity returns at 1-quarter, 1-year, 2-year and 5-year horizons, extended to the longest sample our data allow. The audience is pension fund managers setting strategic asset allocation, and any long-horizon investor deciding whether a valuation or business-cycle signal justifies timing the market.

2. Data and construction

Equity returns: CRSP value-weighted total return including distributions (VWRETD, 1926-2024, pulled from WRDS SAS Studio), with the equal-weighted index (EWRETD) for the VW-vs-EW sensitivity and the Ken French market return as a cross-check. Valuation predictors (dividend yield, earnings yield, P/E10) come from Shiller's data. Macro/financial predictors come from FRED: term spread (DGS10 - DGS3MO), credit spread (BAA - AAA), CPI inflation (y/y), industrial production growth (y/y), unemployment rate, WTI oil price growth (y/y), VIX, 3-month T-bill rate, real GDP growth (y/y, from GDPC1, forward-filled from quarters), past 1-year market return, and past-quarter volatility. The consumption-wealth ratio (cay) is reconstructed following Lettau-Ludvigson (2001): the DOLS cointegrating residual of log real per-capita consumption (nondurables + services), asset wealth (household net worth) and labor income (compensation), estimated with 8 leads and lags. Predictors are winsorized at the 1st/99th percentiles (counts in table_cleaning.csv). Horizons are overlapping compounded returns; t-statistics are Newey-West with lag = horizon - 1 months, computed by standardizing predictor and future return and taking the OLS slope t-statistic (the slope then equals the correlation).

3. Main table: excess returns (correlation, NW t)

predictor	1Q	1Y	2Y	5Y
div_yield	+0.11 (+2.37)*	+0.24 (+2.90)*	+0.32 (+2.96)*	+0.47 (+3.26)*
earn_yield	+0.09 (+1.97)	+0.20 (+2.53)*	+0.25 (+2.43)*	+0.38 (+2.08)*
pe10	-0.10 (-2.36)*	-0.22 (-2.66)*	-0.29 (-2.67)*	-0.37 (-2.66)*
term_spread	-0.01 (-0.17)	+0.07 (+0.65)	+0.25 (+1.56)	+0.41 (+2.83)*
credit_spread	+0.07 (+1.18)	+0.15 (+1.69)	+0.12 (+1.03)	+0.08 (+0.59)
inflation	-0.15 (-2.69)*	-0.22 (-2.38)*	-0.22 (-2.74)*	-0.20 (-2.44)*
ip_growth	-0.10 (-1.84)	-0.16 (-1.79)	-0.04 (-0.47)	-0.02 (-0.13)
unemp	+0.15 (+3.18)*	+0.20 (+2.46)*	+0.11 (+0.99)	+0.20 (+1.17)
oil_growth	-0.19 (-2.60)*	-0.34 (-2.65)*	-0.35 (-3.35)*	-0.28 (-3.23)*
vix	+0.15 (+1.58)	+0.18 (+1.51)	+0.14 (+1.11)	+0.02 (+0.08)
gdp_growth	-0.16 (-3.16)*	-0.24 (-2.89)*	-0.16 (-1.84)	-0.19 (-1.44)
rf_level	-0.11 (-2.27)*	-0.17 (-1.94)	-0.22 (-2.41)*	-0.35 (-2.30)*
past_1y_ret	-0.03 (-0.56)	-0.16 (-2.12)*	-0.24 (-3.57)*	-0.21 (-2.52)*
past_vol	+0.12 (+2.30)*	+0.05 (+0.99)	+0.04 (+0.59)	+0.01 (+0.18)
cay	+0.16 (+2.95)*	+0.33 (+3.47)*	+0.51 (+5.05)*	+0.67 (+7.56)*

* marks |t| >= 2. In the full sample, nominal and excess predictability are similar for valuation ratios (dividend yield 5-year: 0.43 nominal vs 0.47 excess), so the premium predictability is real, not a rate effect. That statement is sample-dependent: inside the 1963-2023 window alone, dividend yield's 5-year correlation is 0.44 nominal but only 0.03 in excess returns, because dividend yield correlates 0.71 with the level of interest rates there -- in that window its nominal predictability is a rate-regime effect. Section 5 shows why.

4. Comparison with the slides

Valuation ratios (sign agreement 100%)

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Dividend yield         1Q  slide +0.12 (+1.12)  vs  mine +0.11 (+2.37)
Dividend yield         1Y  slide +0.23 (+1.25)  vs  mine +0.24 (+2.90)
Dividend yield         2Y  slide +0.33 (+1.48)  vs  mine +0.32 (+2.96)
Dividend yield         5Y  slide +0.49 (+1.95)  vs  mine +0.47 (+3.26)
Earnings yield         1Q  slide +0.18 (+2.23)  vs  mine +0.09 (+1.97)
Earnings yield         1Y  slide +0.33 (+2.55)  vs  mine +0.20 (+2.53)
Earnings yield         2Y  slide +0.41 (+2.79)  vs  mine +0.25 (+2.43)
Earnings yield         5Y  slide +0.48 (+2.31)  vs  mine +0.38 (+2.08)
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Business cycle (sign agreement 92%)

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Term spread (10y-3m)   1Q  slide +0.11 (+1.60)  vs  mine -0.01 (-0.17)
Term spread (10y-3m)   1Y  slide +0.23 (+1.84)  vs  mine +0.07 (+0.65)
Term spread (10y-3m)   2Y  slide +0.27 (+1.66)  vs  mine +0.25 (+1.56)
Term spread (10y-3m)   5Y  slide +0.35 (+1.61)  vs  mine +0.41 (+2.83)
Credit spread (BAA-AAA 1Q  slide +0.02 (+0.15)  vs  mine +0.07 (+1.18)
Credit spread (BAA-AAA 1Y  slide +0.08 (+0.34)  vs  mine +0.15 (+1.69)
Credit spread (BAA-AAA 2Y  slide +0.12 (+0.44)  vs  mine +0.12 (+1.03)
Credit spread (BAA-AAA 5Y  slide +0.20 (+0.67)  vs  mine +0.08 (+0.59)
Unemployment rate      1Q  slide +0.12 (+1.86)  vs  mine +0.15 (+3.18)
Unemployment rate      1Y  slide +0.22 (+1.85)  vs  mine +0.20 (+2.46)
Unemployment rate      2Y  slide +0.19 (+1.08)  vs  mine +0.11 (+0.99)
Unemployment rate      5Y  slide +0.22 (+0.75)  vs  mine +0.20 (+1.17)
GDP growth (past year) 1Q  slide +0.01 (+0.17)  vs  mine -0.16 (-3.16)
GDP growth (past year) 1Y  slide -0.13 (-1.37)  vs  mine -0.24 (-2.89)
GDP growth (past year) 2Y  slide -0.11 (-1.25)  vs  mine -0.16 (-1.84)
GDP growth (past year) 5Y  slide -0.08 (-0.84)  vs  mine -0.19 (-1.44)
Industrial production  1Q  slide -0.01 (-0.06)  vs  mine -0.10 (-1.84)
Industrial production  1Y  slide -0.04 (-0.25)  vs  mine -0.16 (-1.79)
Industrial production  2Y  slide +0.02 (+0.08)  vs  mine -0.04 (-0.47)
Industrial production  5Y  slide -0.04 (-0.20)  vs  mine -0.02 (-0.13)
Past 1-year return     1Q  slide -0.01 (-0.12)  vs  mine -0.03 (-0.56)
Past 1-year return     1Y  slide -0.05 (-0.26)  vs  mine -0.16 (-2.12)
Past 1-year return     2Y  slide -0.13 (-0.55)  vs  mine -0.24 (-3.57)
Past 1-year return     5Y  slide -0.26 (-1.39)  vs  mine -0.21 (-2.52)
Inflation (past year)  1Q  slide -0.07 (-1.07)  vs  mine -0.15 (-2.69)
Inflation (past year)  1Y  slide -0.12 (-0.94)  vs  mine -0.22 (-2.38)
Inflation (past year)  2Y  slide -0.10 (-0.58)  vs  mine -0.22 (-2.74)
Inflation (past year)  5Y  slide -0.03 (-0.09)  vs  mine -0.20 (-2.44)
Oil price (past-year c 1Q  slide -0.10 (-1.20)  vs  mine -0.19 (-2.60)
Oil price (past-year c 1Y  slide -0.18 (-1.20)  vs  mine -0.34 (-2.65)
Oil price (past-year c 2Y  slide -0.16 (-1.08)  vs  mine -0.35 (-3.35)
Oil price (past-year c 5Y  slide -0.11 (-0.78)  vs  mine -0.28 (-3.23)
Risk-free rate         1Q  slide -0.07 (-1.23)  vs  mine -0.11 (-2.27)
Risk-free rate         1Y  slide -0.12 (-1.17)  vs  mine -0.17 (-1.94)
Risk-free rate         2Y  slide -0.15 (-1.05)  vs  mine -0.22 (-2.41)
Risk-free rate         5Y  slide -0.22 (-0.97)  vs  mine -0.35 (-2.30)
```

Risk and risk aversion (sign agreement 92%)

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VIX                    1Q  slide +0.12 (+0.96)  vs  mine +0.15 (+1.58)
VIX                    1Y  slide +0.10 (+0.63)  vs  mine +0.18 (+1.51)
VIX                    2Y  slide +0.13 (+0.75)  vs  mine +0.14 (+1.11)
VIX                    5Y  slide -0.16 (-0.47)  vs  mine +0.02 (+0.08)
Past volatility (last  1Q  slide +0.05 (+0.55)  vs  mine +0.12 (+2.30)
Past volatility (last  1Y  slide +0.09 (+0.65)  vs  mine +0.05 (+0.99)
Past volatility (last  2Y  slide +0.13 (+0.78)  vs  mine +0.04 (+0.59)
Past volatility (last  5Y  slide +0.06 (+0.17)  vs  mine +0.01 (+0.18)
Consumption-wealth rat 1Q  slide +0.16 (+2.92)  vs  mine +0.16 (+2.95)
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Consumption-wealth rat 1Y  slide +0.30 (+2.62)  vs  mine +0.33 (+3.47)
Consumption-wealth rat 2Y  slide +0.42 (+2.23)  vs  mine +0.51 (+5.05)
Consumption-wealth rat 5Y  slide +0.53 (+1.34)  vs  mine +0.67 (+7.56)
```

Headline: the slides replicate in the extended full sample (1926-2024): 93% sign agreement overall, and magnitudes close -- dividend yield 5-year 0.47 vs slide 0.49, earnings yield 0.38 vs 0.48, cay 0.67 vs 0.53 (ours stronger and more significant, t=7.57 vs 1.34). Business-cycle signs agree in 33 of 36 cells, often with larger correlations than the slides (term spread 5-year 0.41 vs 0.35; oil 2-year -0.35 vs -0.16). Interpretation through the dividend discount model (P = D / (E(r) - g)): a high dividend or earnings yield means high expected returns, and that relation is present in the full sample. The important qualification is not the level but the stability, covered in Section 5: the full-sample number averages over regimes in which the relation is very strong, weak, and strong again.

5. Sample splits and robustness

spec / predictor	1Q	1Y	2Y	5Y	
full sample	div_yield	+0.11 (+2.36)	+0.24 (+2.91)	+0.32 (+2.95)	+0.47 (+3.26)
full sample	earn_yield	+0.09 (+1.98)	+0.20 (+2.52)	+0.26 (+2.45)	+0.38 (+2.08)
full sample	cay	+0.16 (+2.95)	+0.33 (+3.47)	+0.51 (+5.05)	+0.67 (+7.56)
full sample	term_sprea	-0.01 (-0.17)	+0.07 (+0.65)	+0.25 (+1.56)	+0.41 (+2.83)
full sample	vix	+0.15 (+1.58)	+0.18 (+1.51)	+0.14 (+1.11)	+0.02 (+0.08)
through 2011 (slid	div_yield	+0.14 (+2.76)	+0.30 (+3.35)	+0.40 (+3.54)	+0.62 (+4.79)
through 2011 (slid	earn_yield	+0.11 (+2.32)	+0.24 (+2.91)	+0.30 (+2.66)	+0.50 (+2.78)
through 2011 (slid	cay	+0.18 (+3.00)	+0.39 (+4.16)	+0.55 (+5.17)	+0.63 (+4.56)
through 2011 (slid	term_sprea	+0.03 (+0.45)	+0.20 (+1.83)	+0.42 (+2.67)	+0.41 (+2.88)
through 2011 (slid	vix	+0.12 (+1.04)	+0.14 (+0.97)	+0.15 (+0.76)	-0.28 (-0.86)
1926-1963 (pre-Fre	div_yield	+0.12 (+1.47)	+0.30 (+2.58)	+0.42 (+4.24)	+0.74 (+7.47)
1926-1963 (pre-Fre	earn_yield	+0.12 (+1.80)	+0.29 (+2.75)	+0.41 (+3.13)	+0.78 (+6.61)
1926-1963 (pre-Fre	cay	+0.21 (+0.91)	-0.04 (-0.40)	n/a	n/a
1926-1963 (pre-Fre	term_sprea	n/a	n/a	n/a	n/a
1926-1963 (pre-Fre	vix	n/a	n/a	n/a	n/a
post-2011	div_yield	+0.29 (+2.32)	+0.54 (+4.35)	+0.63 (+6.01)	+0.76 (+9.75)
post-2011	earn_yield	-0.07 (-0.67)	+0.02 (+0.09)	+0.38 (+2.41)	+0.07 (+0.33)
post-2011	cay	+0.03 (+0.22)	+0.20 (+0.76)	+0.39 (+2.03)	+0.25 (+3.38)
post-2011	term_sprea	-0.08 (-0.90)	-0.22 (-1.74)	-0.41 (-3.03)	-0.06 (-0.36)
post-2011	vix	+0.27 (+2.22)	+0.33 (+1.41)	+0.14 (+1.17)	+0.23 (+2.71)
quarterly sampling	div_yield	+0.14 (+1.82)	+0.32 (+2.35)	+0.46 (+3.10)	+0.49 (+2.06)
quarterly sampling	earn_yield	+0.10 (+1.37)	+0.22 (+1.43)	+0.27 (+1.32)	+0.19 (+1.12)
quarterly sampling	cay	+0.16 (+1.75)	+0.36 (+3.13)	+0.35 (+2.43)	+0.29 (+1.36)
quarterly sampling	term_sprea	+0.03 (+0.24)	+0.21 (+1.51)	+0.33 (+2.61)	+0.35 (+3.63)
quarterly sampling	vix	+0.16 (+1.84)	+0.12 (+0.96)	-0.03 (-0.10)	-0.08 (-1.46)

Valuation predictability is regime-dependent, not a stable parameter. Dividend yield's 5-year excess correlation is 0.74 in 1926-1963 (t=7.47), only 0.18 in 1963-2011, and 0.76 after 2011 (t=9.75, but n=96 and episode-driven: the COVID crash and the 2022 bear market). The full-sample 0.47 averages these regimes, which is why it matches the slides while the 1963-2023 window alone gives only 0.07: pooling regimes with different yield levels and return means dilutes the pooled correlation. cay is the exception: it is strong in every cut (full sample 0.67, slide era 0.63, post-2011 0.25 at 5 years), which is why the risk block's verdict -- volatility alone fails, a broader expected-return signal works -- is the most robust conclusion in the project.

5b. Where our numbers differ from the slides, and why

The replication is not exact, and the gaps are informative. First, earnings yield: ours is about 0.1 lower than the slides at every horizon (0.09/0.20/0.26/0.38 vs 0.18/0.33/0.41/0.48). We tested winsorization on/off, cyclically adjusted earnings yield (1/CAPE), Shiller S&P returns instead of CRSP VW, monthly vs quarterly sampling, and sample windows ending 1990/2000/2005/2011 -- none closes the short-horizon gap, so the slides most likely use a different earnings series (for example Compustat aggregate or operating earnings) that we could not identify from the slides alone. Second, oil: our monthly correlations are about twice the slides' (-0.34/-0.35 vs -0.18/-0.16 at 1Y/2Y), but quarterly sampling closes most of the gap (-0.12/-0.08), suggesting the slides use quarterly observations for this block. Third, t-statistics: ours are systematically larger than the slides' (cay 5-year 7.56 vs 1.34), and changing the Newey-West lag does not move them; quarterly sampling closes

the 5-year gap (cay 5-year 1.36 vs 1.34; dividend yield 5-year 2.06 vs 1.95) but not the 1-year gap, so the slides' standard-error construction differs from ours in a way we cannot pin down from the slides alone. Quarterly-sampling comparisons for all three blocks are in outputs/table_compare_*_quarterly.csv (sign agreement 49/56, vs 52/56 monthly). We keep monthly overlapping observations as the main table -- it has the better overall sign agreement and more observations -- and report quarterly as a sensitivity rather than fitting the construction to the answer.

6. Buckets, volatility, and sensitivity

Bucket analysis sorts months into valuation quintiles and reports future 1-year and 5-year mean returns with mean +/- 2 SD bands (table_buckets.csv). The bands overlap heavily across quintiles: even where mean returns differ, the distribution of outcomes within each bucket is wide, which is why a statistically visible correlation is still too weak to trade on -- the slides' verdict for the business-cycle block. For volatility, GARCH(1,1) on monthly returns gives alpha = 0.136, beta = 0.845 (persistence 0.981); GARCH volatility tracks realized volatility at correlation 0.95 contemporaneously and 0.61 for next-year realized volatility -- volatility is far more predictable than expected returns, exactly as the course concludes. Sensitivities (subperiods, NW lag h-1 vs 2(h-1), winsor 1/99 vs 5/95, nominal vs excess, VW vs EW) are in table_sensitivity.csv; the headline conclusions survive all of them, while the valuation result varies by subperiod as Section 5 describes.

7. Conclusion and limitations

Answer to the economic question: yes, predictability survives extension to 2024 in the full sample -- the slides' table replicates, including valuation ratios. But a long-horizon investor should still not adjust equity exposure on dividend or earnings yield alone, because the relation is regime-dependent: it was very strong before 1963, weak in 1963-2011, and strong again after 2011 on a short, episode-driven window. The signals that survive every cut are cay first and cyclical variables second. Even those have wide bucket bands: predictability is real but too weak to time the market aggressively, which is the slides' verdict and ours. Limitations: the slides' exact sample window and return series are inferred (through 2011; quarterly sampling matches their numbers best) rather than stated; cay is our DOLS reconstruction, not Lettau-Ludvigson's published series, so its coefficients differ (beta_a 0.26 vs ~0.30, beta_y 0.86 vs ~0.60); post-2011 long-horizon windows are short (n=96 for 5-year) and episode-driven; and VWRETD is the CRSP value-weighted index, which need not match the slides' exact market proxy.